



Assessment of Public Information Understanding using Plain Language and Data Visualization

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ABSTRACT

Understanding is an essential quality in the presentation of information, mainly to promote democracy, transparency, and citizen participation in open government data (OGD). Data visualizations are common in open data portals and public services, but there is still a lack of consolidated practices for their publication and evaluations aimed at understanding users. Above all, add the Plain Language technique, which helps to make information more understandable for everyone and reduce barriers to the inclusion of a poorly literate public. This research evaluated the impact of the practices of this language gathered in a usage Guide for creating graphics understandable to the public in relation to other visualizations. The study was carried out with users of public health visualization in the context of the COVID-19 pandemic. The results showed that the use of the Guide improved understanding in tasks with fewer variables and data involved. An important insight for the development of dashboards and public portals with visualizations accessible to all. However, with challenges to be overcome mainly in the size and complexity of the various practices to be followed, making it difficult for users to fully use the Guide. Investing in researching a user support system can ensure higher quality graphics.

CCS CONCEPTS

• Human-centered computing; • Visualization; • Visualization design and evaluation methods;

KEYWORDS

Plain Language, Data Visualization, Open Government Data, Assessment

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1 INTRODUCTION

A world saturated with data demands different qualities in the presentation of information. Access, ease of use, proper organization, among many other essential data qualities [1]. However, one of them becomes paramount and without it the others would be of no use. We're talking about understanding here. Ensuring that those interested in information understand what is transmitted is the basis for everything else. In the public context, this need increases, as we are talking about all citizens who, in order to exercise their citizenship, need to use information. More and more public data are available in open data portals [2], digitization of services, public information dissemination on social media [3] etc. For such resources, understanding is essential in strengthening democracy, increasing transparency and expanding participation, scenarios where the citizen is not broadly inserted [4].

The search for better understanding is also relevant for the inclusion of the public with little (or no) literacy. In Brazil, these rates are high, with 63% of the population below the proficient or intermediate classification for the ability to understand simple information [5]. In this sense, the Plain Language technique can contribute to this topic through a set of good practices for intralingual translation, which consists of interpreting verbal signs through other signs of the same language to facilitate the cognitive process [6]. A general definition, plain language seeks to present the text, structure, and design of information so clearly that the reader can easily find what he is looking for, understand what he found and use that information [7]. However, despite this technique being widely used in government information from several countries [8, 9], it does not have specific guidelines for content other than textual, mainly visual representations such as graphics or visualizations [10].

Data visualizations are widely used in dashboards in general, which in turn are common in the area of open government data (OGD) and public service portals aimed at decision-making and presenting information to citizens [11]. However, examples of incorrect and confusing information are also common, making life difficult for those who need the information [12]. The area of information visualization deals with this aspect, however there is still a lack of consolidated practices on how data should be published for public consumption being fully understood [13, 14] and usability assessments including user studies focusing on understanding [15]. Therefore, this research aims to evaluate the impact of simple language together with the visualization of information in the design of graphics that are more understandable to citizens. A user study was carried out based on public health visualizations in the context of the new coronavirus pandemic, a scenario that further encouraged the use of graphics to disseminate public health information

[16]. Based on graphics created using a Guide that gathered simple language practices, we defined the following objectives:

- Demonstrate that graphs created with the support of plain language practices for visualization are perceived as **more understandable compared to other graphs**, through questions about the data for readers;
- Attest that the use of plain language techniques for visualizations facilitate the creation of more efficient visualizations, in this case with the **potential to be more understandable for presenting information**;

This work presents in the next section a background of works related to the theme. In section 3 we present the practices of the simple language for building graphs, elaborated in a usage guide. In the following section, we have a brief description of the methodology used for the evaluation design. Then, in section 5, we report the execution of the evaluation with the readers of the visualizations and the results found. In section 6, we present the discussions of the study's findings and end with the conclusions, limitations, and future work in section 7.

2 BACKGROUND

Open government data (OGD), open data portals, dashboards, indexes, and evaluation rankings are common in the public context, often used by governments for decision-making and to empower citizens to demand more clarity on topics of greater social interest [17]. In the context of the new coronavirus pandemic, it is no different. Panels with visualizations emerged to disseminate COVID-19 data worldwide [18]. In Brazil, we highlight the index of levels of transparency on public information developed by Open Knowledge Brazil [19] or the ranking of fiscal transparency also in the fight against COVID-19 by Transparency International Brazil [20]. Both assess the quality of data and information related to the pandemic published by the Brazilian federal government and by the states in their official portals. They are relevant, as they seek to minimize and identify some problems involving the data. However, none of them delve into the topic of visualization, assessing whether it is in line with good practices in the area. In the case of open data portals, research by Barcellos, Bernardini and Viterbo [2] highlights that the use of plain language for OGD can be an efficient alternative to promote greater comprehensibility. However, currently these portals do not implement enough computational tools to support the citizen efficiently in interpreting the available data. Ansari, Barati and Martin [15] also reinforce that there is a critical research gap, as we have several case studies describing visualization tools for public data. However, there is a lack of studies with the user's perspective and with a focus on understanding. Using well-designed dashboards can help citizens engage more with OGD. Especially if they guarantee the proper design principles [21]. However, the work by Valotto et al. [22] who evaluated the availability and main characteristics of Brazilian subnational government service portals. One of the main suggestions for improving the functionality of portals involves the difficulty of understanding users, which is one of the barriers to improving the efficiency of public management with digital solutions for citizens.

3 THE GUIDE TO PLAIN LANGUAGE GRAPHICS

To start the evaluation of the use of plain language practices in graphics, a Guide was created to orient the design of graphics, elaborated from the practices of information visualization collected during an initial phase of the research through a systematic review of the literature on the theme [23]. These selected practices are not intended to represent all good practices in the area in a complete way, but to form a minimal initial set that offers a more efficient design for the most common types of graphs (bars, lines, pie, or maps). The set constituted has a total of 55 practices involving various aspects of a visualization, such as colors, legends, graph types, axes, etc [24]. In the Guide, we unite and classify these practices and, to facilitate their use, we translate the textual description of all guidelines into plain language, in order to avoid ambiguities and difficulty in interpreting them. In addition, we organized the Guide, with several practical examples, as shown in Figure 1.

4 ASSESSMENT DESIGN APPROACH

The evaluation of the artifacts of this research is closely linked to the subject of understandability, comprehensibility and clarity, terms that are highly subjective, which can vary their interpretation according to the reader's knowledge, available resources, environment variables, among others. However, in the area of visualization, some works provide evaluation methods for these cases that can help to reduce subjectivity. Objectively measuring the quality of a solution created by a participant can be difficult if the solution is not easily quantifiable and can only be judged subjectively [25]. An alternative is to use an evaluation standard known as "Human Black Box", initially used in the work of [26], where the authors add a second phase to the initial experiment, in which the participants perform several graphic tasks using the views generated by users from a first stage. In other words, instead of trying to qualitatively evaluate the artifact, during the second phase we measured the readers' performance through questionnaires on the graphs produced by the initial experiment, as shown in Figure 2. With this, from these responses, we will have metrics on the quality and understandability of each graph created by users in phase one.

In phase one, visualizations were created by 4 non-specialist developers who work or have worked in public bodies in Brazil. Two of these developers (users A and B) received dummy data about COVID-19, specific tasks presented below, and the practice Guide, and the other two just the data and tasks (users C and D):

- **Task 1:** analyze and demonstrate the total number of deaths from COVID-19 in each Brazilian state. Objective: the reader should be able to observe which states have the highest and lowest number of deaths.
- **Task 2:** analyze and represent the total number of vaccines in Brazil in relation to the country's population. Objective: the reader should be able to observe how close or not vaccination is to reaching the entire population of Brazil.
- **Task 3:** analyze and demonstrate the evolution of hospitalizations due to COVID-19 in the first 8 months of the year (January to August) in the state of Mato Grosso (MT).



Figure 1: Guide to Plain Language Charts example pages.

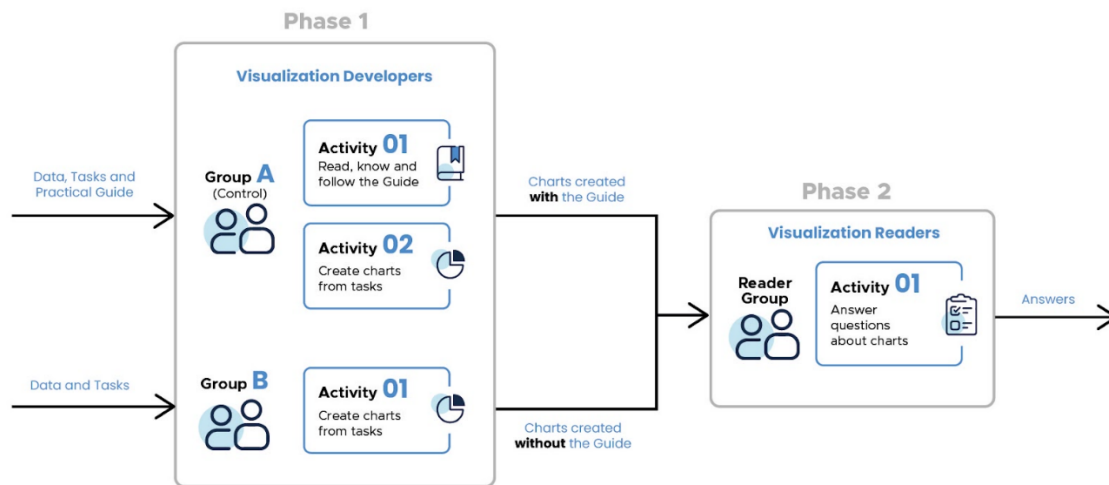


Figure 2: Two-stage evaluation methodology.

Objective: the reader should be able to observe and understand the variation in the number of hospitalizations in the represented period.

- **Task 4:** analyze and represent the number of hospitalizations due to COVID-19 during the months of February to July in the state of Pernambuco (PE) in comparison to the number of hospitalizations due to COVID-19 during the same period in the state of Espírito Santo (ES), Paraná (PR), Rondônia (RO) and Tocantins (TO). Objective: the reader should be able to observe and understand the variation in the number of hospitalizations between states in the represented period.

Each participant in this phase developed one or more charts for each task asynchronously and freely, a sample of the charts is shown in Figure 3. After that, the charts produced by each group were collected and phase two of evaluation with the readers was started on online questionnaires. The questionnaires were prepared

from questions associated with pairs of graphs, one of them created with the guidelines of the Guide and the other without them. In each pair, one or more questions are asked related to the task and objective that the graphics in question should meet according to the model of user tasks proposed by Munzner [27]. For example, for task 1, the reader is asked which states have the highest and lowest number of deaths? The participant provides an answer by analyzing the two images of the charts: chart A (created with the support of the Guide), chart B (created without the Guide). Next, ask yourself which of the graphs did the participant find easier to understand, helping you with the previous questions? Three response alternatives are provided, where only one can be chosen: graph A, graph B or both graphs. Finally, an open field for insertion of comments by the questionnaire respondent is also available for a subsequent analysis of the reasons for selecting the readers in each case. This question structure is followed by the four tasks and their respective graphs created by the participants in the first phase. In

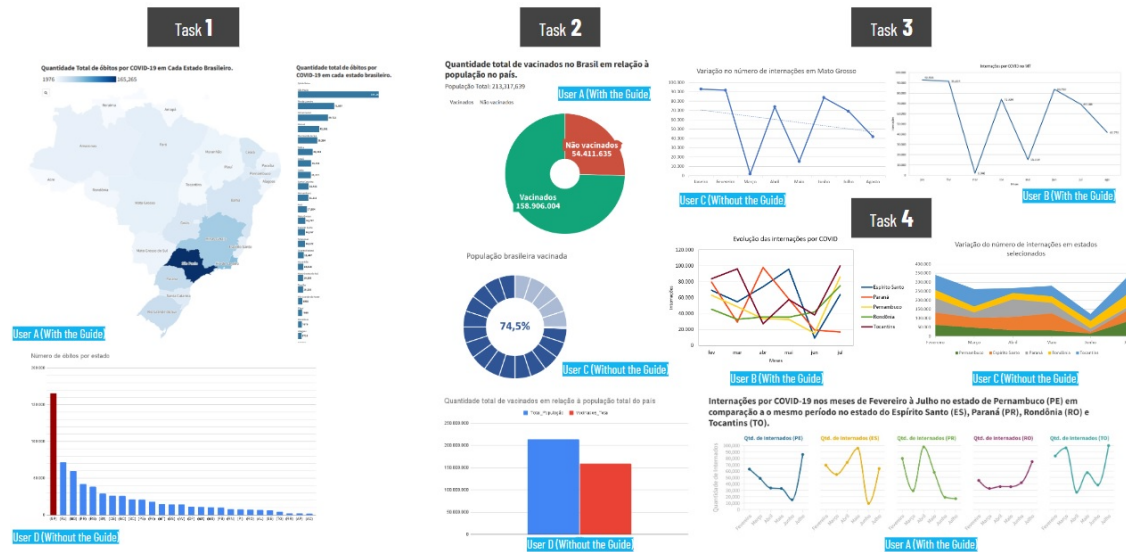


Figure 3: Examples of graphics generated during phase 1 of the evaluation.

these questions, the data collected reflects the number of statements that associate graph A or B as clearer, or else, in the case of the number of statements with the option of “both graphs”, the lack of distinction between them, where respondents do not perceive differences between the cases. Finally, the last questions in the questionnaire are the characterization of the respondent’s profile such as age, gender, education, location, and level of knowledge in graph interpretation varying on a 10-point scale (0 associated with “lack of knowledge in graph interpretation” up to the value 10 representing “total experience in the subject”).

5 EVALUATING WITH READERS FROM DATA VISUALIZATIONS

The questionnaires begin with a brief explanation of the research objectives, the anonymity of the collected data and a confirmation of consent to proceed with the answers. They were divided so that the questionnaire respondents had access to a single graph for each group in the previous phase, that is, with a graph generated with the support of the Guide and another not in each task. A raffle defined that the graphs of user A would be in the questionnaire next to that of user D. Likewise, the graphs of users B and C were included together in another form. This assessment was shared on the researcher’s social media and others were asked to pass on the research (snowballing method) between December 2022 and January 2023. In total, there were 112 responses received between the questionnaires (60 responses in questionnaire 1 on graphs from user B and C and 52 responses in questionnaire 2 on user charts A and D).

In questionnaire 1, 32 (53%) participants are men and 28 are women (47%). Most are between 21 and 30 years old (37%), followed by a portion aged 41 to 50 years old (25%). The 31 to 40 age group was in third place with a total of 10 respondents (17%). Between 51 and 60 years old were 12% of the respondents (7 participants), with only 5% between 18 and 20 years old and 5% over 60 years

old. The predominant level of education was postgraduate, whether masters, doctors or with specialization (53%). About 33% have higher education and another 13% are in high school. The vast majority resides in Rio de Janeiro in Brazil (65%) and the second-largest public comes from another Brazilian state, Minas Gerais (13%). Participants from other regions of the country were also present, but in amounts smaller than 5%. There were even 2 participants from abroad. When asked about their knowledge of interpreting and reading graphs on a scale from 0 to 10, most respondents categorized themselves as level 7 (25%) and 8 (25%), followed by 20% identifying themselves as level 9 of knowledge. Participants who identified themselves as level 5 were only 13%. The highest level (10) achieved adherence by 7% of participants. At lower levels such as 1 to 4 just over 4% identified themselves as lay at that level.

In questionnaire 2 we obtained a total of 52 participants, of which 33 are women (63%) and 19 are men (36%). Most are between the ages of 21 and 30 (33%), followed by 29% between their 41 and 50s. The age group of 31 to 40 years old was in 3rd place with a total of 21% of adhesion, followed by 10% between 51 and 60 years old. Only 4% are over 60 years old and in the ranges between 18 and 20 years old and under 18 years old we have 2% in each. As for the level of education of the participants in this questionnaire, 56% have a master’s, doctorate or specialization. 31% are in higher education (undergraduate) while 13% in high school. Again, the state of Rio de Janeiro had the highest number of participants, followed by a tie between São Paulo, Minas Gerais and Paraná with 3 participants each (6% in each state). Other states do not exceed 4% (2 participants) and there was no participation of people from outside Brazil. Regarding the level of knowledge for interpreting and reading graphs (scale from 0 to 10), most participants in this group, 40% (21 participants) identified with level 8. 17% adhered to level 7, followed by level 6 or 10 with 11% each. Level 9 obtained 10% and the lowest levels, from 5 to 1, did not exceed 4% each.

Table 1: Data from phase 2 assessment questionnaires with visualization readers.

QUESTIONNAIRE 1	User B	User C	-
Task 1	Charts WITH the Guide	Charts WITH the Guide	Both Charts
Correct answer	13	24	22
Incorrect answer	0	0	1
Total	13	24	23
Task 2			
Correct answer	13	23	7
Incorrect answer	5	9	3
Total	18	32	10
Task 3			
Correct answer	13	12	3
Incorrect answer	15	6	11
Total	28	18	14
Task 4			
Correct answer	14	33	2
Incorrect answer	4	5	2
Total	18	38	4
TOTAL PARTICIPANTS	60		
QUESTIONNAIRE 2	User A	User D	-
Task 1	Charts WITH the Guide	Charts WITH the Guide	Both Charts
Correct answer	15	19	18
Incorrect answer	0	0	0
Total	15	19	18
Task 2			
Correct answer	33	8	6
Incorrect answer	4	0	1
Total	37	8	7
Task 3			
Correct answer	12	7	7
Incorrect answer	8	10	8
Total	20	17	15
Task 4			
Correct answer	12	25	5
Incorrect answer	6	3	1
Total	18	28	6
TOTAL PARTICIPANTS	52		

Table 1 presents the data obtained by both questionnaires and their respective answers in each task and groups of graphs.

6 DISCUSSIONS

In each task, respondents should indicate which graph they found easier to understand, helping them with questions about the data. It is noted that the question involving the graph of task 3 was considered more difficult among users, given the significant increase in wrong answers in this item (58 errors in 112 answers). In addition, the data help us to observe in which tasks the graphs generated with the help of the Guide can be considered more assertive. The graphs in Figure 4 demonstrate this topic, considering only the data from the respondents who got the questions right, thus considering that the graph helped them in some way in the correct answer.

In task 1, we observed that the group of graphics created without the support of the Guide fared better, with about 39% choosing it as “the easiest to understand” of those who answered the questions about these graphics correctly in both questionnaires. Still, the difference is close to those who selected both groups of charts as easy, considering that there were no aspects that made the charts better than others.

In the scenario of task 2, we see that the graphs created with the support of the Guide were considered easier to understand the data and support the solution of the task. A percentage of 51% of respondents considering this group with the best performance is displayed. It is also notable that compared to the previous task, the category in which the “two groups” are considered comprehensible received the lowest adherence rate, only 15% in this case.

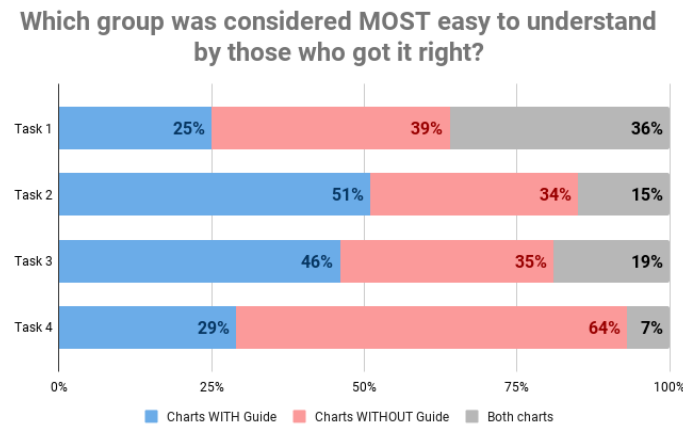


Figure 4: Analysis of responses to both questionnaires considering only data from readers who answered the questions correctly.

Observing task 3, the scenario is repeated. The graphs created with the Guide are perceived as easier for this activity among the participants who got the asked questions right (46%). However, the difference this time is smaller, only 11 points, and there is even an increase in the group where the two graphs are understood as easy, with about 19% in this group.

Finally, during task 4 the opposite of the previous situations occurs. In this case, graphics built without support from the Guide are considered easier to understand. Figure 4 shows that more than twice the percentage of graphics chosen with the Guide (29%) were chosen without the Guide (64%). This task was the one in which the most disrespected practices were observed by one of the graphs created with the Guide, which may help explain the reason for this result.

During data analysis, the total percentages of readers were also considered, that is, considering both those who got the questions right and those who missed the questions in each task. However, the data did not change significantly, with graphics created with the Guide remaining the most selected in tasks 2 and 3. In task 1, only 25% of all participants preferred the graphics created with the Guide, equal to the result when considered only the readers who got it right, as shown in Figure 4. In tasks 2 and 3, the result was also similar when we analyzed all the participants and not just those who got it right, 49% and 43% of preference for the Guide's graphics respectively. The result is also similar in task 4, however, with a slightly larger margin for the Guide's graphics. Of the total participants, 32% preferred the graphics developed with the Guide, as opposed to the 29% of those who answered the question correctly. In addition, we observed whether performance had improved considering only the data from questionnaire 2, with graphs made by users A and D. Since user B, despite having the Guide, reported that he did not use it completely and did not finish reading it completely. However, the data remained similar to the graphs generated, adding all the participants who answered correctly in the two questionnaires. Only in tasks 2 and 3 that the graphics made with the support of the Guide were considered easier in this scenario.

7 CONCLUSIONS

In this two-phase experiment, we aimed to demonstrate that the Guide can help create more understandable graphics in general. The results showed that tasks 2 and 3 were perceived by more users as being better understood due to the graphics being created using the Guide. Graphs created with the Guide were perceived as more understandable in tasks with fewer variables and data involved. The rates found in the tasks performed using the Guide were very close to those performed without using it, mainly because the users who had access to the Guide did not use the Guide's guidelines in their entirety. This may mean that although the Guide is considered useful and relevant, the effectiveness of its use still needs to be guaranteed. A point perhaps to be considered is the large number of practices to be observed when drawing up the graphs. We consider that this may be an indication that investing in research on the user support computational system in the verification, guidance, and compliance with the practices of the Guide, as fundamental to guarantee higher quality graphics.

The research was limited to presenting and validating the artifacts associated with contexts related to health at the governmental level, it is necessary to advance in other scenarios to prove the generalization of the artifacts produced. The results obtained by the evaluations carried out during the research provide a demonstration of the contribution of the Guide and its practices, however, they were limited to a few participants (only 4) in phase 1 of the evaluation. Another point is that we found little use of the Guide in the development of graphics, a broader study with more engaged users may provide more results of the Guide's effectiveness. In addition, the results obtained were evaluated from questions about the data, new studies with more possibilities of tasks or analysis of graphs can generate interesting results.

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